GE Medical Systems

Technical Publication

Direction 2301701-100
SnapShot Segment Image Quality

OPERATORS DOCUMENTATION

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Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

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REVISON HISTORY

Revision	Date	Reason for change
0	5/16/01	SnapShot Segment Image Quality

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INTRODUCTION

When scanning a Cardiac exam using your LightSpeed Plus scanner there are potential image artifacts that may occur. This document will inform the user of the image quality expected when imaging the heart, and also the potential artifacts that may occur.

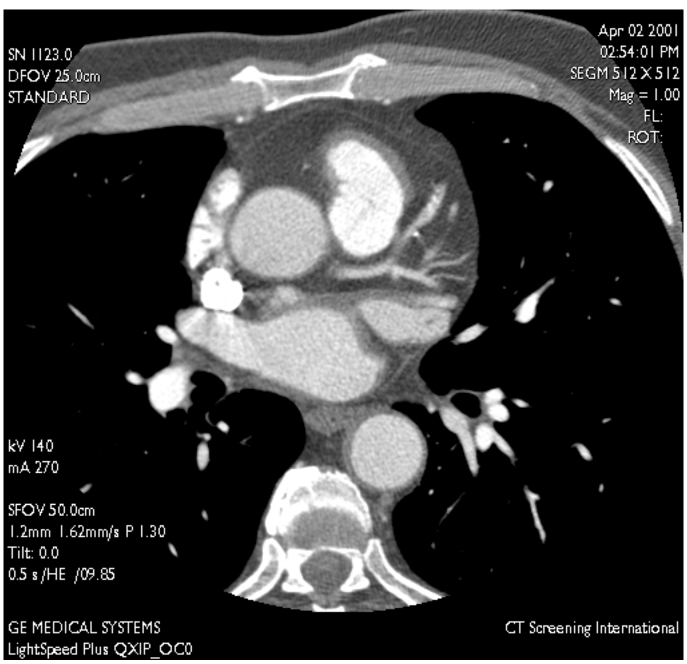
Some of the potential artifacts are:

- Blurred Coronary Arteries
- Missing RCA or Black Circular Artifact
- Superior Vena Cava(SVC) Streaking in 2D and Banding Artifacts in 3D images
- Phase Misregistration in 2D and 3D Images

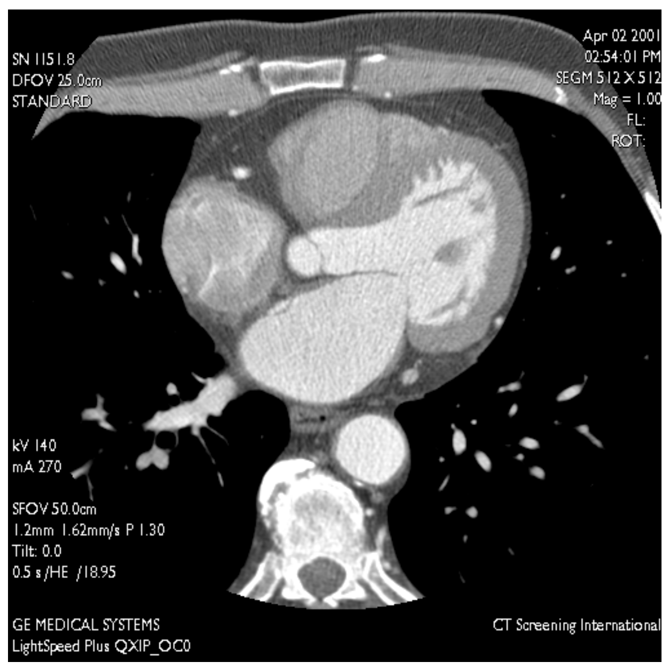
CARDIAC IMAGES USING SNAPSHOT SEGMENT

When scanning a cardiac exam it is important to follow the recommend guidelines in the LightSpeed Plus operator manual. When these guidelines are followed, the user can expect cardiac images with motion free vessels and no severe artifacts. The following two images are examples of cardiac images where the recommend guide lines were followed.

Axial LAD & LCx

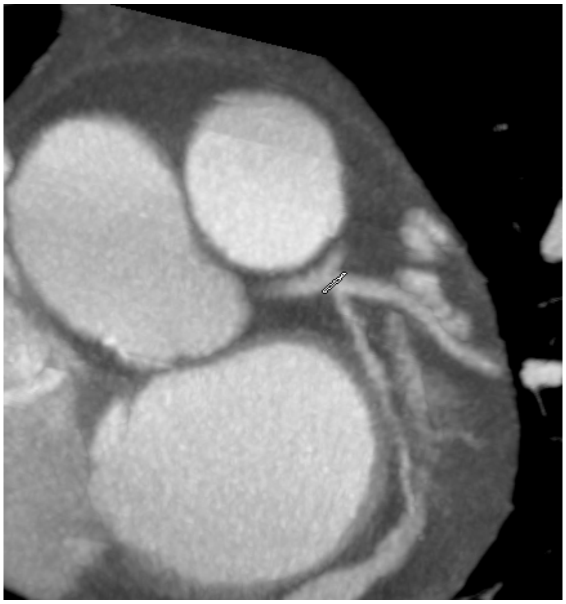


Axial RCA

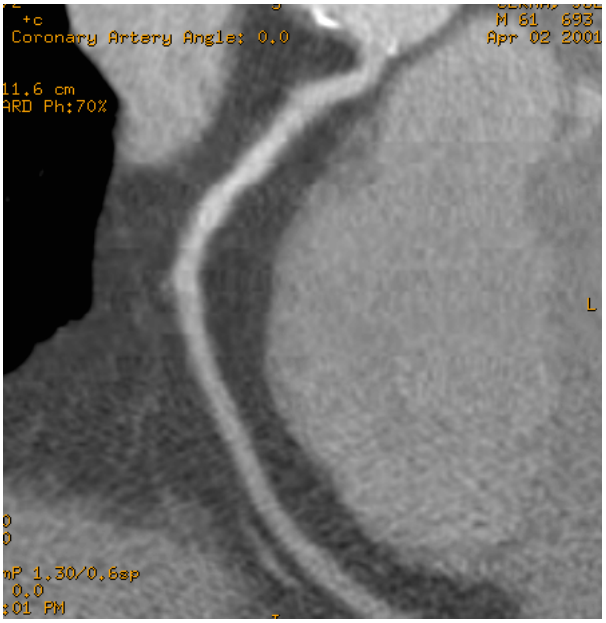


Curved Reformat of the LAD

Oblique Reformat of the LCX & Marginal Branch



Curved Reformat of the RCA



Note: Whenever viewing images using a post processing techniuqe, ie vessel analysis, always refer to the 2D axial image if there is an area of suspission.

BLURRED CORONARY ARTERIES IN 2D IMAGES

In 2D images the coronary artery can look blurred due to inadequate temporal resolution or patient motion. This is usually caused by scanning a patient with a higher heart rate than is recommend (75bpm or below) or when a patient is breathing during the scan.

To avoid blurred arteries, avoid scanning a patient with a heart rate above 75bpm. Care should also be taken to explain the exam to the patient and inform them of the length of the breath hold. If the breath hold is too long, inform the patient to slowly release their breath at the end of the exam.

If a patient has been scanned with a heart rate below 75bpm and the images are still blurry try to produce the images at a different phase location. Sometimes this will reduce the blurring in the vessels. The right coronary artery is especially prone this artifact.

Blurred LCA and LCx

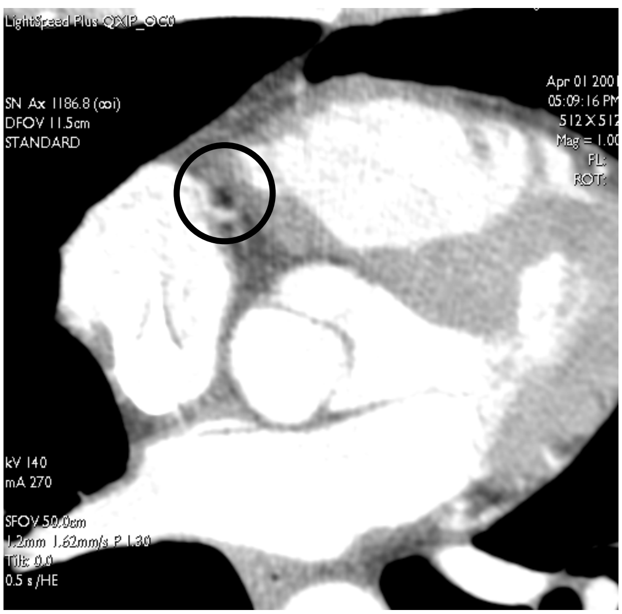
Blurred LCA and LCx

Blurred/Whispy RCA

MISSING RCA OR BLACK CIRCULAR ARTIFACT

In some 2D images of the coronary artery the right coronary artery may disappear or a black circular artifact may appear where the RCA normally lies. The black circular artifact may also appear in the 3D images. This is due to insufficient temporal resolution, which in turn will not freeze the motion of the RCA.

The RCA is the most difficult artery to image due to the motion of the artery in the diastolic phase. It is thought that the artery is best seen in end systolic phase, so if the right coronary artery is missing in some of the images or this circular artifact appears, the user can try and reconstruct the images in another phase.



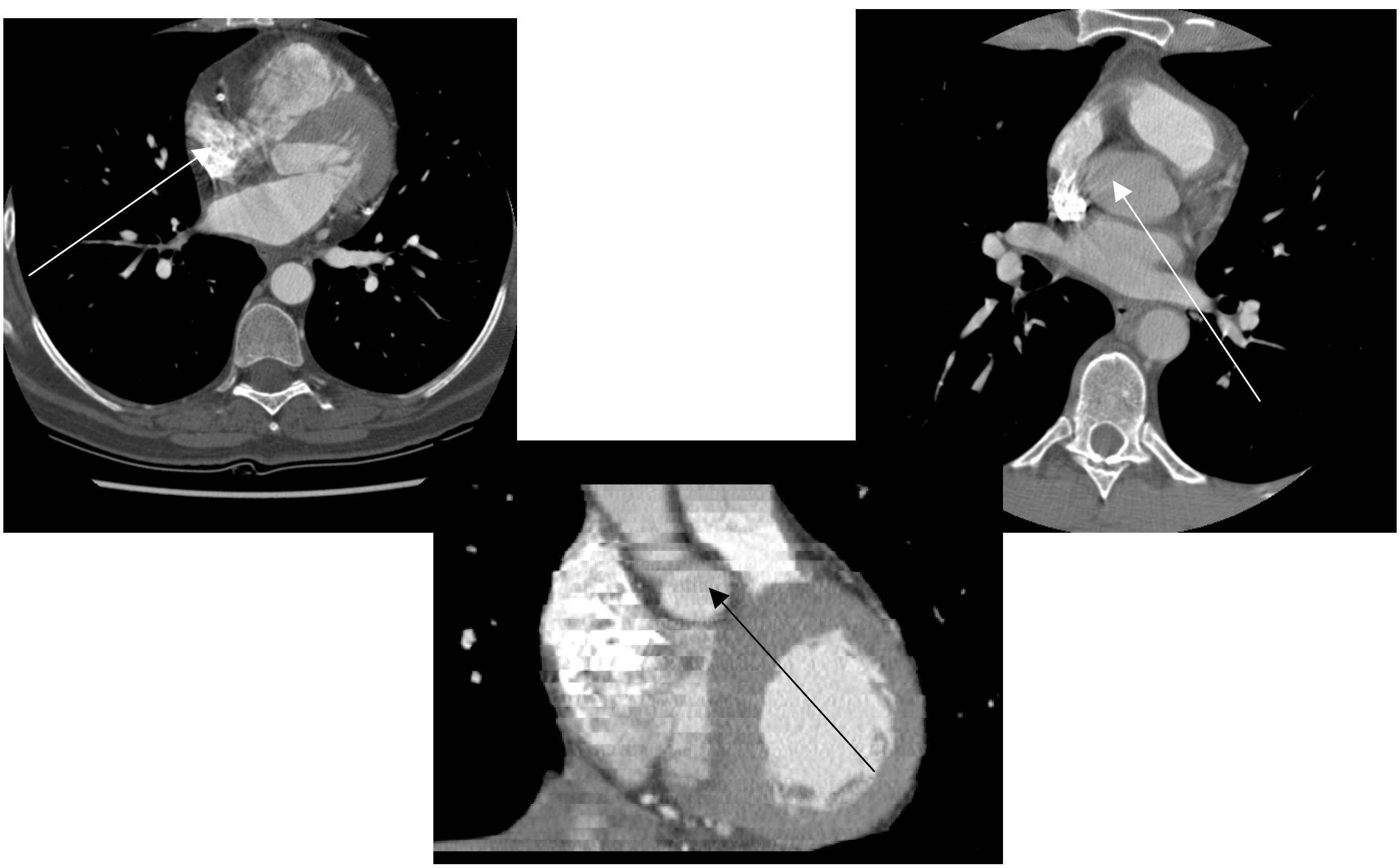
SUPERIOR VENA CAVA (SVC) STREAKING IN 2D AND BANDING ARTIFACTS IN 3D IMAGES

In 2D images of the heart the user may notice streaking artifact in the area of the Superior Vena Cava, this is due to the large amount of contrast flowing through the superior vena cava.

In the 3D images of the heart the user will notice banding artifacts in the sagittal and coronal images. This is due to CT number shifts or noise pattern differences between horizontal bands/regions on the images. Each band corresponds to a set of axial images generated from one cardiac cycle. The following may be causes for banding artifacts:

- Contrast non-uniformity - for contrast enhanced tissue who's CT number varies with time, CT number differences between heart cycles will be accentuated by gated reconstruction.
- SVC and metal streaks – streaking due to high contrast in the SVC can affect the CT numbers of surrounding tissue, when viewed as reformat the user may see gray scale shifting from band to band
- Noise patterns – for larger patients noise patterns in the axial images may be more apparent in reformatted images from band to band.

The SVC artifact can be reduced by scanning inferior to superior, but there is a risk that the patient may not be able to hold their breath during the complete scan, which in-turn will cause motion artifacts in the proximal coronary arteries.



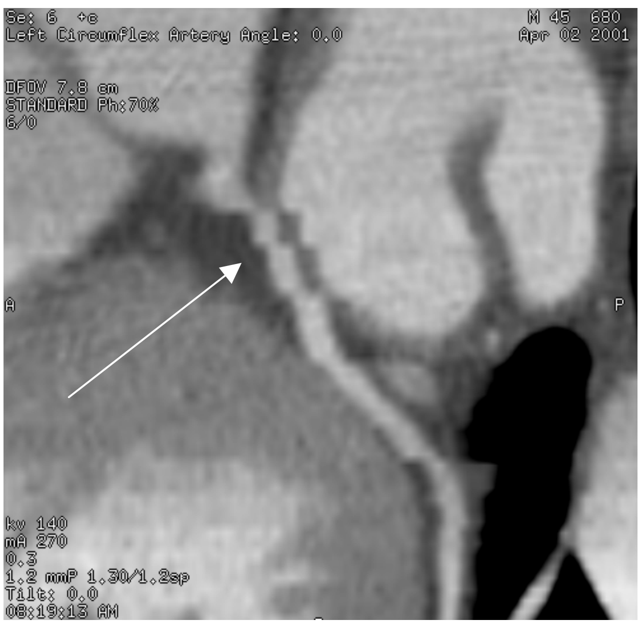
PHASE MISREGISTRATION IN 2D AND 3D IMAGES

In the 2D images of the heart the user will notice an apparent "beating" of the heart as they scroll through a series of the cardiac axial images. When the user pages through the images they will notice the artery/cardiac structures moving from side to side or up and down.

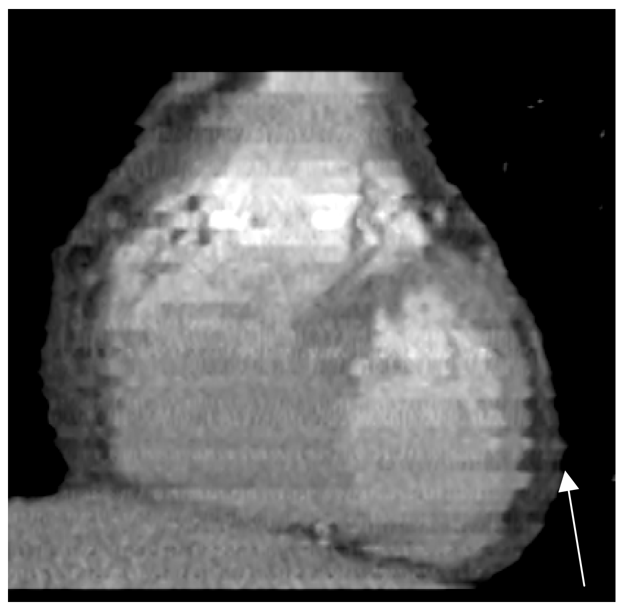
In the 3D images the heart will display horizontal shifting in the coronal/sagittal images where the heart could not be frozen at the same position or phase location from heart cycle to heart cycle. This artifact is most prominent in the mid section of the heart and is visible at the edge of the left ventricle.

Phase misregistration may be caused by inadequate temporal resolution, which in turn does not freeze the motion of the heart in patients with high heart rates. It is recommend scanning patients with heart rates below 75bpm. Phase misregistration may also be caused by the ventricle contracting differently from cycle to cycle (image to image) causing the artery or location of the ventricle to change from one image to another causing misregistration.

LCX with Misregistration



Short Axis with Misregistration



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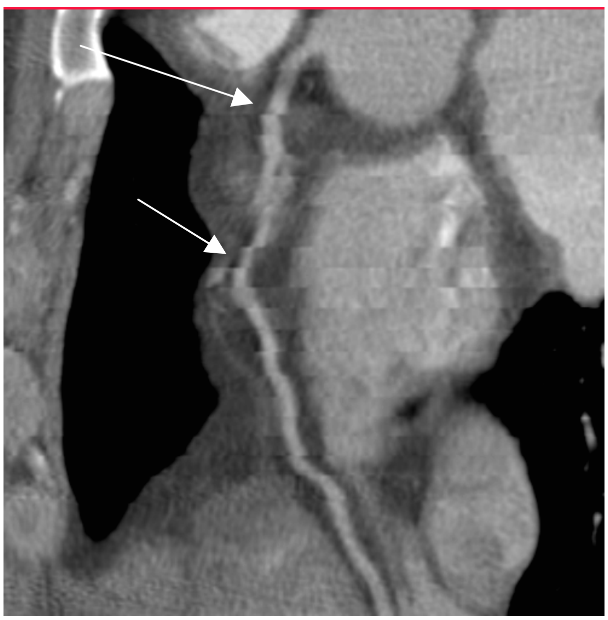
PHASE MISREGISTRATION IN VESSEL ANALYSIS

When using the Vessel Analysis software the user may create an image and notice the vessel has misregistration artifacts. This artifact may be corrected by producing an image in another phase and then running vessel analysis again. As you can see from the case below, the first RCA artery was produced with a 70% phase location and throughout the data set the artery has misregistration artifacts. When the user changes the phase location to 80% most of the artifacts disappeared.

Misregistration 70% R to R Interval



Improvement with 80% of R to R Interval

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